Adaptive-Scale Group Bipartite Consensus of Second-Order Multi-Agent Systems with Uncertain Parameters

1st Yuhan Peng
School of Mathematics and Statistics
Donghua University
Shanghai, China
yuhan_peng@outlook.com

2nd Xinyan Yang
School of Mathematics and Statistics
Donghua University
Shanghai, China
xinyan_yang0110@outlook.com

3rd Youzhi Sun
School of Mathematics and Statistics
Donghua University
Shanghai, China
youzhi_sun@outlook.com

***Abstract*—This paper studies adaptive-scale group bipartite consensus for second-order multi-agent systems with unknown nonlinear dynamics and uncertain parameters. A leader–follower framework is considered over a signed network with both cooperative and antagonistic interactions, where structural balance is used to describe the topology. The followers are partitioned into multiple subgroups with scaling factors introduced to capture heterogeneous responses to the leader. Under strong connectivity and structural balance assumptions, relevant coupling matrices are shown to be nonsingular M-matrices and positive definite. A linearly parameterized model is used to handle unknown nonlinear dynamics. A distributed adaptive control protocol and parameter update law are designed for online compensation. Lyapunov analysis, together with Barbalat's lemma, shows that the closed-loop system achieves group bipartite consensus. Under persistent excitation, parameter estimation errors converge to zero. Simulation results verify the effectiveness and robustness of the proposed method under signed topology and nonlinear uncertainties.**

***Index Terms*—Second-order multi-agent systems, Adaptive-scale group bipartite consensus, Unknown nonlinear dynamics, Signed networks, Distributed adaptive control**

I. Introduction

In recent years, the issue of consensus in multi-agent systems (MASs) has attracted the attention of many researchers and has been the subject of in-depth and extensive study. To date, many results have been achieved and applied in various practical fields, such as drone formations, sensor networks, intelligent transportation systems and distributed robots.

Early research focused primarily on first-order integrator models, which assume that the dynamics of each agent are solely determined by the first-order derivative of its own state [11]. This model is simple in structure and easy to analyse, providing a foundation for understanding distributed cooperative behaviour. However, in practice, the motion of agents is typically influenced by both position and velocity, making it difficult to accurately describe their dynamics using only first-order models. To more accurately characterize the motion of real systems, researchers have further extended the consensus problem to second-order MASs [1]. Unlike first-order systems, second-order systems incorporate both position and velocity state variables, enabling a more comprehensive description of the dynamics of physical systems such as mobile robots and drone formations. Therefore, the study of consensus problems in second-order MASs holds significant theoretical and engineering value.

As the study of the consensus theory of MASs progresses, the traditional consensus is no longer able to adequately describe coordination activities in the complex network. In order to describe the network interactions that simultaneously exhibit both cooperative-competitive interactions, the issue of consensus in signed networks was studied Altafini in his paper [2]. The author introduced the concept of bipartite consensus. The research shows that when the network satisfies the condition of structural balance, all agents eventually converge to two values with equal magnitudes but opposite signs This allows them to effectively describe the phenomenon of extreme views in the social network and the antagonistic cooperative behavior in cooperative–competitive systems. Since then, the study of bipartite consensus has become an important research area in the field of MASss and has received a lot of attention.

It worth noting that in many real-world scenarios, the agents is not simply divided into two opposing groups, but rather exhibits a more complex group structure. In response to this issue, Liu et al. proposed the concept of Group bipartite consensus (GBC) in the literature [3]. This concept integrates both group consensus and bipartite consensus, achieving a cooperative-antagonistic evolutionary behavior among multiple sub-groups. In comparison to traditional bipartite consensus, Group bipartite consensus can more accurately describe the complex network structures and group evolutionary laws. Subsequently, Lu and Liu further expanded the application of Group bipartite consensus to include various systems, such as heterogeneous systems, non-linear systems, and second-order multi-agent systems [3] [4].

Although group bipartite consensus can describe multi-group cooperative behavior in complex networks, most existing studies assume that agents ultimately converge to target states with equal absolute values. However, in practical applications such as formation control, cooperative tracking,

and task allocation, different agents often need to complete cooperative tasks according to predetermined proportional relationships—that is, their final states must satisfy specific proportional constraints rather than being exactly identical. To address this need, Roy proposed the concept of scaled consensus in [5] and investigated the conditions for achieving scaled consensus in linear network systems. Compared to traditional consensus, scaled consensus allows each agent to converge according to a predefined proportional relationship, thereby providing a more flexible design framework for the cooperative control of MASs. Building on this, Yu et al. in [6] investigated the problem of scaled group consensus in first- and second-order MASs, establishing the relevant theory for achieving scaled group consensus under directed topologies, which laid the foundation for the application of scaled mechanisms in complex cooperative control tasks.

Current research on group bipartite consensus and scaled consensus is mostly based on the assumption that the system models are precisely known. However, most practical systems exhibit certain nonlinearities and uncertainties. Ignoring these unknown factors may not only degrade the robustness and control performance of the control system, but may also prevent the consensus objective from being achieved.

Existing studies mainly address nonlinear dynamics through boundedness assumptions, Lipschitz conditions, or adaptive approximation methods. However, most of these approaches either require restrictive assumptions or focus on conventional consensus problems, leaving the combination of nonlinear uncertainties and complex cooperative-antagonistic structures insufficiently investigated. Zhao et al. [14] proposed a distributed finite-time consensus protocol based on relative position and velocity measurements, and proved that the system can achieve consensus within finite time when the nonlinear errors between the leader and followers are bounded. Liu et al. [15] investigated the leaderless consensus problem of first-order nonlinear multi-agent systems under directed switching topologies, where the nonlinear terms satisfy the Lipschitz condition. If the unknown nonlinear dynamics of the system satisfy the linear parameterization condition, an adaptive parameter update law can be designed to identify the unknown parameters online. Yu et al. [6] proposed a decentralized adaptive consensus algorithm by combining local consensus errors with relative position feedback, which addressed the leader–follower consensus problem of first-order multi-agent systems under jointly connected topologies. Tu et al. [9] designed a finite-time adaptive parameter update law and consensus protocol using relative position information feedback, solving the adaptive finite-time consensus problem under fixed and switching topologies. Jiang et al. [7] studied the adaptive consensus problem of second-order nonlinear systems with uncertain parameters based on the model reference approach; however, the convergence of parameter estimation errors to zero cannot be guaranteed. Therefore, for second-order nonlinear multi-agent systems with uncertain parameters, simultaneously achieving consensus and system identification remains an important and challenging problem worthy of further investigation.

With regard to system uncertainty, adaptive control has attracted widespread attention due to its ability to compensate for uncertain dynamics through online parameter estimation mechanisms. Existing research indicates that, for nonlinear systems satisfying linear parameterization conditions, online identification and dynamic compensation of system uncertainty can be achieved by constructing a parameter estimator. Jiang et al. investigated the consensus for second-order nonlinear MASs in directed graphs based on a Model Reference Adaptive Control (MRAC) [7]; Tian et al. designed a fully distributed bipartite consensus control protocol for heterogeneous MASs under dynamically switching topologies [8]. While such research ensures that the system achieves consensus objectives, it typically only guarantees that parameter estimation errors are bounded, and it is difficult to further ensure that the estimated parameters converge to the true parameters.

In fact, if an adaptive controller can not only ensure consensus but also achieve asymptotic convergence of the parameter estimation error to zero, it means that the controller has simultaneously accomplished both cooperative control and system identification. To address this issue, Yu et al. designed a distributed adaptive consensus algorithm [6] using local consensus error and relative state feedback information, thereby achieving leader-follower consensus and parameter convergence; Tu et al. studied the problem of adaptive finite-time consensus for uncertain systems under both fixed and switching topologies [9].

On the other hand, persistent excitation (PE) conditions are widely regarded as a key factor in ensuring the convergence of parameter identification and have been successfully applied in the fields of adaptive estimation and system identification [10]. Nevertheless, most existing research on PE conditions has focused on general consensus or cooperative network scenarios; there remains a relative lack of research on the group-bipartite consensus in cooperative-adversarial networks, particularly when this consensus involves scale-dependent characteristics.

Existing research has yielded substantial results in the areas of group bipartite consensus, scaled consensus, and adaptive control, respectively; however, for second-order multi-agent systems with uncertain parameters, the problem of cooperative control that integrates scale mechanisms, group bipartite structures, and parameter identification capabilities has not yet been fully explored. In particular, within a cooperative-adversarial network environment, how to achieve scaled group bipartite consensus while ensuring that uncertain parameters can be accurately estimated and compensated for remains an important topic worthy of in-depth research.

Based on the above analysis, this paper investigates the problem of adaptive-scaled group bipartite consensus of second-order multi-agent systems with uncertain parameters. By constructing a group bipartite consensus protocol with scaling factors, introducing a parameter adaptive estimation mechanism, and utilizing Lyapunov stability theory, we prove that the system can achieve scaled group bipartite consensus.

Building on this, we prove that, under the condition that the regression matrix satisfies the PE condition, the parameter estimation error asymptotically converges to zero, thereby enabling online estimation and precise estimation of the system's uncertain parameters.

The main contributions of this paper are as follows:

1) For second-order multi-agent systems with cooperative-competitive interactions, a scaled group bipartite consensus control framework was developed. Depending on the choice of scale coefficients, the problem can be reduced to a general bipartite consensus problem.
2) To account for uncertain parameters in system dynamics, an adaptive parameter update law based on local consensus error was designed, enabling online estimation and compensation of uncertain system parameters and reducing the controller's reliance on precise model information.
3) Based on information regarding the relative positions and velocities of the agents, a distributed adaptive control protocol was developed that enables the system not only to achieve scaled group bipartite consensus but also to converge the parameter estimation error to zero, provided that the regression matrix satisfies the PE condition.

II. PRELIMINARIES

A. Graph Theory

This paper considers the consensus problem of leader–follower multi-agent systems. The interactions between agents can be conceptualized through an directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$, where $\mathcal{V} = \{v_1, v_2, ...v_N\}$ is the set of vertices, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges, and $A = [a_{ij}]_{N\times N} \in \mathbb{R}^{N\times N}$ is the weighted adjacency matrix. a_{ij} denotes the weight of the edge from node v_j to node v_i. If $(v_j, v_i) \in \mathcal{E}$, then $a_{ij} \neq 0$. Specifically, $a_{ij} \geq 0$ indicates a cooperative interaction between agents v_j and v_i, whereas $a_{ij} \leq 0$ indicates an antagonistic interaction. Otherwise, $a_{ij} = 0$. For convenience, let $a_{ii} = 0$. The in-degree of node v_i is defined as $d_i = \sum_{j\in\mathbb{N}_i} |a_{ij}|$, where $\mathbb{N}_i = \{j \in \mathcal{V} | (v_j, v_i) \in \mathcal{E}\}$. The degree matrix is given by $D = diag\{d_1, d_2, ...d_N\}$. Then, the Laplacian matrix of graph is defined as $L = [l_{ij}]_{N\times N}$, where

$$l_{ij} = \begin{cases} \sum\limits_{j\in\mathbb{N}_i} |a_{ij}|, & i = j, \\ -a_{ij}, & i \neq j. \end{cases}$$

Define the matrix $B = diag\{b_1, b_2, ...b_N\}$ to represent the communication between the leader and followers. If follower v_i can receive information from the leader, then $b_i > 0$; otherwise, $b_i = 0$. Let $H = L + B$.

According to their different tasks and roles, the follower agents are further divided into $s(s \geq 2)$ subgroups. The i-th$(i = 1, 2...s)$ subgroup contains N_i agents, and its corresponding topology subgraph is denoted as $\mathcal{G}_i = \{\mathcal{V}_i, \mathcal{E}_i, A_{ii}\}$. A scaling factor c_i is introduced for each subgroup $\mathcal{G}_i$, and the matrix C is defined as follows:

$$C = \begin{bmatrix} c_1 I_{N_1} & 0 & \cdots & 0 \\ 0 & c_2 I_{N_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_s I_{N_s} \end{bmatrix}$$

Definition 1 (Structurally Balanced Graph): A graph $\mathcal{G}$ is said to be structurally balanced if its node set can be exactly partitioned into two antagonistic subgroups, i.e., there exist two node subsets $\mathcal{V}_1, \mathcal{V}_2 \subseteq \mathcal{V}$, satisfying $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$, $\mathcal{V}_1 \cap \mathcal{V}_2 = \emptyset$, such that the edge weights within each subgroup are positive, while the edge weights between different subgroups are negative.

Assumption 1: The graph $\mathcal{G}$ is assumed to be strongly connected and structurally balanced. Moreover, there exists $\exists j \in \{1, \ldots, N\}, \quad b_j > 0$. Each subgroup topology $\mathcal{G}_i = \{\mathcal{V}_i, \mathcal{E}_i, A_{ii}\}$ is strongly connected and structurally balanced, and at least one follower can receive information from the leader.

B. Preliminary Lemmas

Lemma 1 [2]: If $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$ is a structurally balanced graph with the node set partitioned into $\mathcal{V}_1$ and $\mathcal{V}_2$, then there exists a diagonal matrix $\Sigma = diag(\sigma_1, ...\sigma_N)$, where

$$\sigma_j = \begin{cases} 1, & v_j \in V_1 \\ -1, & v_j \in V_2 \end{cases},$$

such that $\Sigma A \Sigma \geq 0$. The matrix Σ is called the signature matrix.

Lemma 2: [12] Let $A \in \mathbb{R}^{n\times n}$ be a nonsingular M-matrix. Then there exist two positive column vectors $x = (x_1, x_2, \ldots, x_n)^T = A^{-1}\mathbf{1}_n$, $y = (y_1, y_2, \ldots, y_n)^T = (A^T)^{-1}\mathbf{1}_n$. Define $W = \text{diag}(y_1/x_1, y_2/x_2, \ldots, y_n/x_n)$, Then both W and $WA + A^T W$ are positive definite.

Lemma 3: [2] Under Assumption 1, $\Sigma H \Sigma$ is a nonsingular M-matrix. Moreover, there exists a positive definite diagonal matrix Q such that $\Gamma = QH + H^T Q$ is positive definite.

Assumption 2: There exists a constant $\alpha > 0$, such that $\alpha\lambda_{min}(\Gamma) > 2\lambda_{max}(Q)$.

C. Problem Formulation

Consider a multi-agent system consisting of N followers and one leader. The dynamics of follower agent i are given by

$$\begin{cases} \dot{x}_i(t) = v_i(t), \\ \dot{v}_i(t) = u_i(t) + f_i(x_i(t), v_i(t)), \end{cases} \tag{1}$$

where $x_i(t), v_i(t) \in \mathbb{R}$ denote the position and velocity information of agent i at time t, respectively, and $u_i(t) \in \mathbb{R}$ represents the control input. $f_i(x_i(t), v_i(t))$ is an unknown

continuous nonlinear function representing the unknown dynamics of the system, which can be transformed into the following linearly parameterized form:

$$f_i(x_i(t), v_i(t)) = \phi_i^T(x_i(t), v_i(t))\theta_i \tag{2}$$

where $\phi_i \in \mathbb{R}^m$ denotes the known basis function vector, which is determined by the system state and velocity, and $\theta_i \in \mathbb{R}^m$ represents the unknown parameter vector.

The dynamics of the leader are given by

$$\begin{cases} \dot{x}_0(t) = v_0(t), \\ \\ \dot{v}_0(t) = u_0(t), \end{cases} \tag{3}$$

where $x_0(t), v_0(t) \in \mathbb{R}$ denote the position and velocity of the leader at time t, respectively. $u_0(t) \in \mathbb{R}$ represents the control input, which is assumed to be bounded.

Definition 2: For the multi-agent systems in Eq.(1) and Eq.(3), if for any initial conditions, the following holds:

$$\begin{cases} \lim_{t\to+\infty} \|x_i(t) - c_k\sigma_i x_0\| = 0, \\ \\ \lim_{t\to+\infty} \|v_i(t) - c_k\sigma_i v_0\| = 0, \end{cases} \tag{4}$$

where $i \in \mathcal{V}_k$,$k = 1, 2, ...s$. Then, the system is said to achieve scale-group bipartite consensus.

III. MAIN RESULTS

To solve the scale-group bipartite consensus problem described in Eq.(4), the following control protocol is designed for $i \in \mathcal{V}_k$, $k = 1, 2, ...s$:

$$\begin{aligned} u_i = & -\alpha \left(\sum_{j\in\mathbb{N}_i} |a_{ij}| \left(x_i - \mathrm{sgn}(a_{ij})x_j\right) + b_i \left(x_i - c_k\sigma_i x_0\right) \right) \\ & -\alpha \left(\sum_{j\in\mathbb{N}_i} |a_{ij}| \left(v_i - \mathrm{sgn}(a_{ij})v_j\right) + b_i \left(v_i - c_k\sigma_i v_0\right) \right) \\ & -\phi_i^T\hat{\theta}_i + c_k\sigma_i u_0. \end{aligned} \tag{5}$$

where α is a parameter to be selected, and $\hat{\theta}_i$ denotes the estimate of the unknown parameter θ_i. The variables ξ_i, η_i are defined as the local consensus errors:

$$\begin{cases} \xi_i = \sum_{j\in\mathbb{N}_i} |a_{ij}|(x_i - \mathrm{sgn}(a_{ij})x_j) + b_i(x_i - c_k\sigma_i x_0) \\ \eta_i = \sum_{j\in\mathbb{N}_i} |a_{ij}|(v_i - \mathrm{sgn}(a_{ij})v_j) + b_i(v_i - c_k\sigma_i v_0). \end{cases} \tag{6}$$

The adaptive parameter update law is designed as follows:

$$\dot{\hat{\theta}}_i = \phi_i \sum_{j=1}^{N} h_{ji} q_j (\xi_j + \eta_j), \tag{7}$$

Let $\hat{\Theta} = (\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_N)$, then the system control law is given by:

$$\begin{cases} U = -\alpha\xi - \alpha\eta - \Phi^T\hat{\Theta} + u_0 C\Sigma\mathbf{1}_N \\ \\ \dot{\hat{\Theta}} = \Phi H^T Q(\xi + \eta). \end{cases} \tag{8}$$

where $U = (u_1, u_2, \ldots, u_N)^T$, $\xi = (\xi_1, \xi_2 \cdots \xi_N)^T, \eta = (\eta_1, \eta_2 \cdots \eta_N)^T$.

To prove that the proposed consensus protocol enables the system to achieve scale-group bipartite consensus, the tracking error is first defined as follows:

$$e_{xi} = x_i - c_k\sigma_i x_0, \quad e_{vi} = v_i - c_k\sigma_i v_0. \tag{9}$$

Accordingly, define $e_x = (e_{x1}, e_{x2}, \ldots, e_{xN})^T$ and $e_v = (e_{v1}, e_{v2}, \ldots, e_{vN})^T$. For $i \in \mathcal{V}_k$, $k = 1, 2, ...s$.

Based on this, the local consensus errors can be rewritten in matrix form as follows:

$$\begin{cases} \xi = He_x \\ \eta = He_v. \end{cases} \tag{10}$$

Substituting Eq.(1) and Eq.(8) into Eq.(10), we obtain

$$\begin{cases} \dot{\xi} = \eta, \\ \dot{\eta} = H\left(-\alpha\xi - \alpha\eta + \Phi^T\bar{\Theta}\right)), \end{cases} \tag{11}$$

where $\Phi = diag(\phi_1, \phi_2, \ldots, \phi_N)$, $\Theta = (\theta_1, \theta_2, \ldots, \theta_N)^T$. Define the parameter estimation error as $\bar{\theta}_i = \theta_i - \hat{\theta}_i$, and let $\bar{\Theta} = (\bar{\theta}_1, \bar{\theta}_2, \ldots, \bar{\theta}_N)$. Then the error system can be obtained as

$$\begin{cases} \dot{\xi} = \eta, \\ \dot{\eta} = H\left(-\alpha\xi - \alpha\eta + \Phi^T\bar{\Theta}\right), \\ \dot{\bar{\Theta}} = -\Phi H^T Q(\xi + \eta). \end{cases} \tag{12}$$

Theorem 1: The MAS in Eq.(1) satisfying Assumptions 1–2 can achieve scale-group bipartite consensus under the control protocol in Eq.(8). *Proof:* To prove Theorem 1, the following lemma is first introduced.

Lemma 4: [13] If the function $\dot{f}(t)$ is uniformly continuous and satisfies $\int_0^\infty f(t)dt < \infty$, then $\lim_{t\to\infty} f(t) = 0$.

Eq.(4) requires that $\lim_{t\to+\infty} \|e_x\| = 0$ and $\lim_{t\to+\infty} \|e_v\| = 0$. Since H is invertible, it is sufficient to prove that $\lim_{t\to+\infty} \|\xi\| = 0 \lim_{t\to+\infty} \|\eta\| = 0$.

Consider the following Lyapunov function:

$$V = \frac{1}{2}[\xi^T, \eta^T] \begin{bmatrix} b\Gamma & Q \\ \\ Q & Q \end{bmatrix} \begin{bmatrix} \xi \\ \\ \eta \end{bmatrix} + \frac{1}{2}\bar{\Theta}^T\bar{\Theta} \tag{13}$$

It can be easily verified that V is positive definite, and $V = 0$ if and only if $\|\xi\| = 0, \|\eta\| = 0, \|\bar{\Theta}\| = 0$. Taking the derivative of V along the trajectories of the error system in Eq. (error), we obtain:

$$\begin{aligned} \dot{V} &= [\xi^T, \eta^T] \begin{bmatrix} \alpha\Gamma & Q \\ Q & Q \end{bmatrix} \begin{bmatrix} \dot{\xi} \\ \dot{\eta} \end{bmatrix} + \bar{\Theta}^T\dot{\bar{\Theta}} \\ &= \alpha\xi^T\Gamma\eta + \eta^T Q\eta + \xi^T Q\dot{\eta} + \eta^T Q\dot{\eta} + \bar{\Theta}^T\dot{\bar{\Theta}} \end{aligned}$$

Define $N = \xi^T Q\dot{\eta} + \eta^T Q\dot{\eta} + \bar{\Theta}^T\dot{\bar{\Theta}}$. Substituting Eq. (12) into the above equation yields:

then

$$\dot{V} = -\frac{\alpha}{2}\xi^T\Gamma\xi - \eta^T\left(\frac{\alpha}{2}\Gamma - Q\right)\eta, \tag{14}$$

which implies $\dot{V} \leq 0$. Therefore, $V(\infty)$ exists and is bounded. Furthermore, we obtain:

$$\begin{aligned}\int_0^{\infty} \dot{V}(s)\,\mathrm{d}s &= \lim_{t\to\infty}\int_0^t \dot{V}(s)\,\mathrm{d}s \\ &= \lim_{t\to\infty}\int_0^t \left[-\frac{\alpha}{2}\xi^T\Gamma\xi - \eta^T\left(\frac{\alpha}{2}\Gamma - Q\right)\eta\right]\mathrm{d}s \\ &= V(\infty) - V(0) < \infty.\end{aligned}$$

By Assumption 2 and Lemma 4, $\lim_{t\to+\infty}[-\frac{\alpha}{2}\xi^T\Gamma\xi - \eta^T(\frac{\alpha}{2}\Gamma - Q)\eta] = 0$. Since Γ and $\frac{\alpha}{2}\Gamma - Q$ are both positive definite matrices, it follows that $\lim_{t\to+\infty}\|\xi\| = 0$, $\lim_{t\to+\infty}\|\eta\| = 0$. Therefore, Theorem 1 is proved.

Assumption 3: For the regressor vector $\phi(t)$, there exist constants $T > 0a > 0$, such that, for any t,

$$\int_t^{t+T} \phi(\tau)\phi^T(\tau)d\tau \geq \alpha I \tag{15}$$

Theorem 2: Under Assumption 3, the parameter estimation error of the system in Eq. (1) converges to zero, i.e., $\lim_{t\to+\infty}\|\bar{\Theta}\| = 0$. *Proof:* From Eq. (12), we have

$$\ddot{\eta} = H(-a\ddot{\xi} - a\dot{\eta} + \Phi^T\dot{\bar{\Theta}} + \dot{\Phi}^T\bar{\Theta}). \tag{16}$$

Since $\dot{\xi}, \dot{\eta}$ and $\dot{\bar{\Theta}}$ are bounded, and Φ and $\dot{\Phi}$ are uniformly bounded, it follows that $\ddot{\eta}$ is bounded, i.e., $\ddot{\eta} \in L_\infty$. Therefore, $\dot{\eta}$ is uniformly continuous. According to Lemma 4, $\lim_{t\to\infty}\dot{\eta}(t) = 0$. Furthermore, we obtain $\lim_{t\to+\infty}\Phi^T\bar{\Theta} = 0$. Thus, $\lim_{t\to+\infty}(\Phi^T\bar{\Theta})^T\Phi^T\bar{\Theta} = \lim_{t\to+\infty}\bar{\Theta}^T\Phi\Phi^T\bar{\Theta} = 0$. Therefore,

$$\lim_{t\to+\infty}\int_t^{t+T}(\Phi^T\bar{\Theta})^T\Phi^T\bar{\Theta}d\tau = \lim_{t\to+\infty}\int_t^{t+T}\bar{\Theta}^T\Phi\Phi^T\bar{\Theta}d\tau = 0. \tag{17}$$

By Assumption 3, we have

$$\int_t^{t+T}\bar{\Theta}^T\Phi\Phi^T\bar{\Theta}ds \geq \alpha\int_t^{t+T}\bar{\Theta}^T\bar{\Theta}ds \geq 0, \tag{18}$$

Hence,

$$\lim_{t\to+\infty}\int_t^{t+\delta_0}\bar{\Theta}^T\bar{\Theta}\,d\tau = 0, \tag{19}$$

which implies $\lim_{t\to+\infty}\|\bar{\Theta}\| = 0$. Therefore, Theorem 2 is proved.

IV. NUMERICAL SIMULATIONS

To verify the effectiveness and correctness of the proposed adaptive-scale group bipartite consensus protocol and its parameter update law in, numerical simulations are conducted on the second-order multi-agent system with unknown nonlinear dynamics. The simulations are implemented on the MATLAB platform, where the continuous-time solver ode45 is adopted. Both the relative and absolute error tolerances are set to 10^{-4}.

Consider a second-order multi-agent network consisting of one leader (labeled as node 1) and $N = 5$ followers (labeled as nodes 2–6). The information interaction among the agents is described by the directed signed communication topology shown in Fig. 1.

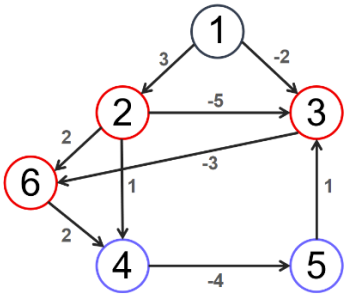


Fig. 1: The network topology.

From Fig. 1, it can be seen that the overall topology is strongly connected and structurally balanced. According to the cooperative (positive-weight) and antagonistic (negative-weight) interactions among the agents, the follower set $\mathcal{V}_F$ is partitioned into two disjoint structurally balanced subsets. Specifically, the first subset, $\mathcal{V}^1 = \{2, 4, 6\}$, is assigned diagonal entries $d_i = 1$, while the second subset, $\mathcal{V}^2 = \{3, 5\}$, is assigned diagonal entries $d_i = -1$. Accordingly, the signature matrix is given by $D = \mathrm{diag}\{1, -1, 1, -1, 1\}$.

The information transmission from the leader to the followers is characterized by the diagonal matrix B. In this example, the leader communicates only with nodes 2 and 3 through directed links. The corresponding coupling weights are 3 and 2, respectively. Therefore, the leader coupling matrix is given by $B = \mathrm{diag}\{3, 2, 0, 0, 0\}$.

In the simulation, the true unknown parameter vector of each agent is set to $\theta_i = [1, 1]^T$, while the initial estimate of the unknown parameter for each follower is uniformly chosen as $\hat{\theta}_i(0) = [0,\ 0]^T$.

The system is simulated over the time interval t[0,60] s. The leader dynamics are given by

$$\begin{cases}\dot{x}_0(t) = v_0(t), \\ \dot{v}_0(t) = -x_0(t).\end{cases}$$

The analytical solution of the above second-order linear ordinary differential equation is

$$\begin{cases}x_0(t) = R\sin(t+\varphi), \\ v_0(t) = R\cos(t+\varphi),\end{cases}$$

where the amplitude R and the phase angle φ are uniquely determined by the initial state of the leader. The initial position and velocity are chosen as $x_0(0) = 0.5$, and $v_0(0) = 0.5$, respectively. Eliminating the time variable t yields the following relationship in the state space $x_0^2(t) + v_0^2(t) = R^2$. Therefore, the leader follows a circular trajectory with radius R in the phase plane. Accordingly, the followers in the cooperative subgroup $\mathcal{V}^1$ asymptotically track the original trajectory of the leader, i.e., $\lim_{t\to\infty} x_i(t) = -x_0(t)$. As a result, all agents in the multi-agent network eventually converge to and evolve on the same limit cycle in the phase plane.

After transforming the nonlinear dynamic function of the followers into a linearly parameterized form, the basis function vector is selected as $\phi_i(x_i, v_i) = [\sin(x_i), \cos(v_i)]^T$. The control parameter is set as $\alpha = 1.5$. Therefore, the conditions in Theorems 1 and 2 are satisfied. The initial value of the parameter estimate is set as a two-dimensional zero vector. The initial position and velocity vectors of the followers are respectively given as follows:

$$x(0) = [1.4132, -3.5162, 4.0086, -3.3932, 0.2651]^T$$

$$v(0) = [4.1792, -3.2310, -0.3752, 3.2345, 0.1431]^T$$

The numerical simulation results are shown as follows.

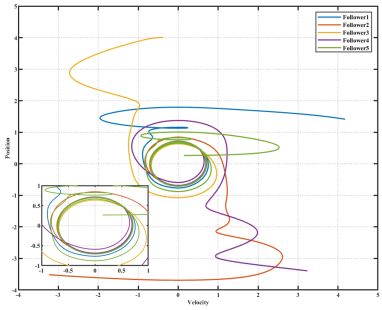


Fig. 2: Spiral phase plane.

The simulation results verify the effectiveness of the proposed protocol.

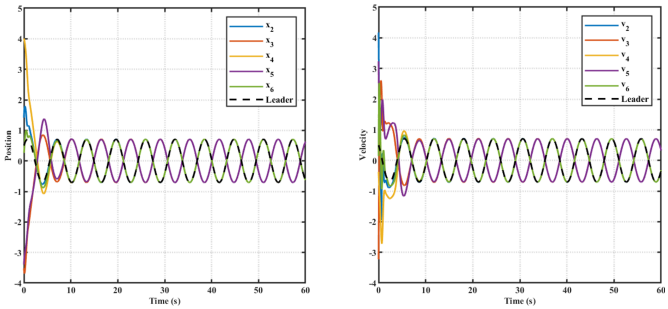
V. CONCLUSION

This paper investigates the adaptive-scale group bipartite consensus problem for second-order multi-agent systems with unknown nonlinear dynamics and uncertain parameters. First, a scale-group bipartite consensus framework is established for leader–follower multi-agent systems over signed communication networks with cooperative and antagonistic interactions. Scaling factors are introduced to characterize the heterogeneous responses of different subgroups to the leader. Then, a distributed adaptive consensus control protocol and a parameter update law are designed to compensate for the unknown nonlinear dynamics online, and sufficient conditions for achieving scale-group bipartite consensus are derived. Furthermore, by constructing a Lyapunov function and applying Barbalat's lemma, it is rigorously proved that the proposed control protocol enables the closed-loop system to achieve scale-group bipartite consensus. Under the persistent excitation condition, the parameter estimation errors are further shown to asymptotically converge to zero. Finally, numerical simulations are provided to verify the effectiveness and robustness of the proposed method.

REFERENCES

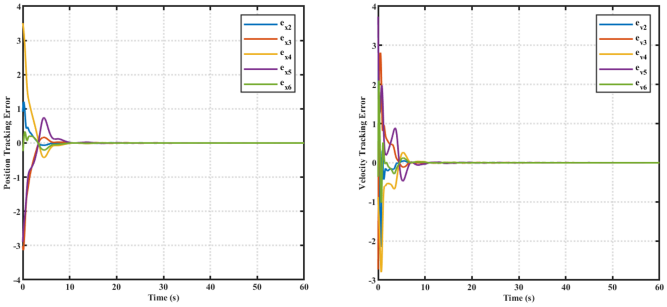
[1] W. Yu, G. Chen, and M. Cao, "Some necessary and sufficient conditions for second-order consensus in multi-agent dynamical systems," *Automatica*, vol. 46, no. 6, pp. 1089–1095, Jun. 2010.

[2] C. Altafini, "Consensus problems on networks with antagonistic interactions," *IEEE Trans. Autom. Control*, vol. 58, no. 4, pp. 935–946, Apr. 2013.



(a) Position trajectories. (b) Velocity trajectories.

Fig. 3: State trajectories of all agents.



(a) Position error. (b) Velocity error.

Fig. 4: State error of all agents.

[3] J. Liu, H. Li, J. Ji, and J. Luo, "Group-bipartite consensus in the networks with cooperative-competitive interactions," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 67, no. 12, pp. 3292–3296, Dec. 2020.

[4] R. Lu, J. Wu, X. Zhan, and H. Yan, "Finite-time group-bipartite consensus tracking for second-order nonlinear multi-agent systems," *Neurocomputing*, vol. 545, Art. no. 126283, 2023.

[5] S. Roy, "Scaled consensus," *Automatica*, vol. 51, pp. 259–262, Jan. 2015.

[6] H. Yu and X. H. Xia, "Adaptive consensus of multi-agents in networks with jointly connected topologies," *Automatica*, vol. 48, no. 8, pp. 1783–1790, Aug. 2012.

[7] C. Jiang and J. Mei, "Adaptive consensus for second-order uncertain multi-agent systems under a directed graph," in *Proc. Chinese Control Decis. Conf. (CCDC)*, 2018, pp. 1335–1340.

[8] C. C. Tian, K. X. Tian, J. Mei, and Y. Wang, "Bipartite leaderless position consensus of heterogeneous uncertain multi-agent systems under switching directed graphs," *Syst. Control Lett.*, vol. 181, Art. no. 105628, 2023.

[9] Z. Z. Tu, H. Yu, and X. H. Xia, "Decentralized finite-time adaptive consensus of multi-agent systems with fixed and switching network topologies," *Neurocomputing*, vol. 219, pp. 59–67, Jan. 2017.

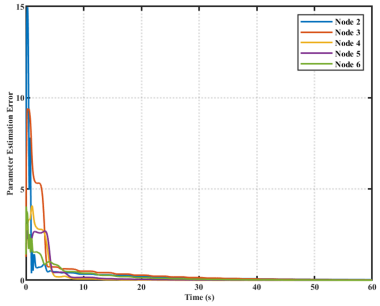


Fig. 5: Parameter estimation error.

[10] E. Nuño, A. Loría, A. I. Paredes, and T. Hernández, "Consensus-based formation control of multiple nonholonomic vehicles under input constraints," *IEEE Control Syst. Lett.*, vol. 6, pp. 2767–2772, 2022.
[11] R. Olfati-Saber and R. M. Murray, "Consensus problems in networks of agents with switching topology and time-delays," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1520–1533, Sep. 2004.
[12] M. Fiedler and V. Pták, "On matrices with non-positive off-diagonal elements and positive principal minors," *Czechoslovak Math. J.*, vol. 12, no. 87, pp. 382–400, 1962.
[13] W. Ren and R. W. Beard, "Consensus seeking in multiagent systems under dynamically changing interaction topologies," *IEEE Trans. Autom. Control*, vol. 50, no. 5, pp. 655–661, May 2005.
[14] Y. Zhao, Z. Duan, G. Wen, and G. Chen, "Distributed finite-time tracking of multiple non-identical second-order nonlinear systems with settling time estimation," *Automatica*, vol. 64, pp. 86–93, Feb. 2016.
[15] W. Liu, S. Zhou, Y. Qi, and X. Wu, "Leaderless consensus of multi-agent systems with Lipschitz nonlinear dynamics and switching topologies," *Neurocomputing*, vol. 173, part 3, pp. 1322–1329, Jan. 2016.